AWS SECURITY ARCHITECTURE

Amazon CloudFront Origin Access Control for Private S3

Serving private S3 content through CloudFront, without ever exposing the bucket to the public internet.

PROJECT ID: 24CC3014-P106
TEAM - 388

TEAM S-62-MA

TEAM MEMBER	NUMBER
NEELI KUNDANA VENI	2400030515
T. DEEPAK KUMAR PATRO	2400030499
GRANDHI SUNISHKA	2400030480
SNEHASIVA NAGA SREE VAKKALAGADDA	2400030452

AWS SECURITY ARCHITECTURE

PROBLEM STATEMENT:

Modern applications need to deliver static content such as web pages, images, videos, and files quickly and securely. However, directly exposing an S3 bucket can create security risks.

- Public S3 buckets can expose sensitive or important content.
- Direct access to the S3 origin bypasses centralized access control.
- Delivering content directly from S3 may not provide the performance benefits of a global CDN.
- Data stored in S3 should be protected against unauthorized access through encryption at rest.
- Organizations need a solution that provides secure access without making the S3 bucket public.

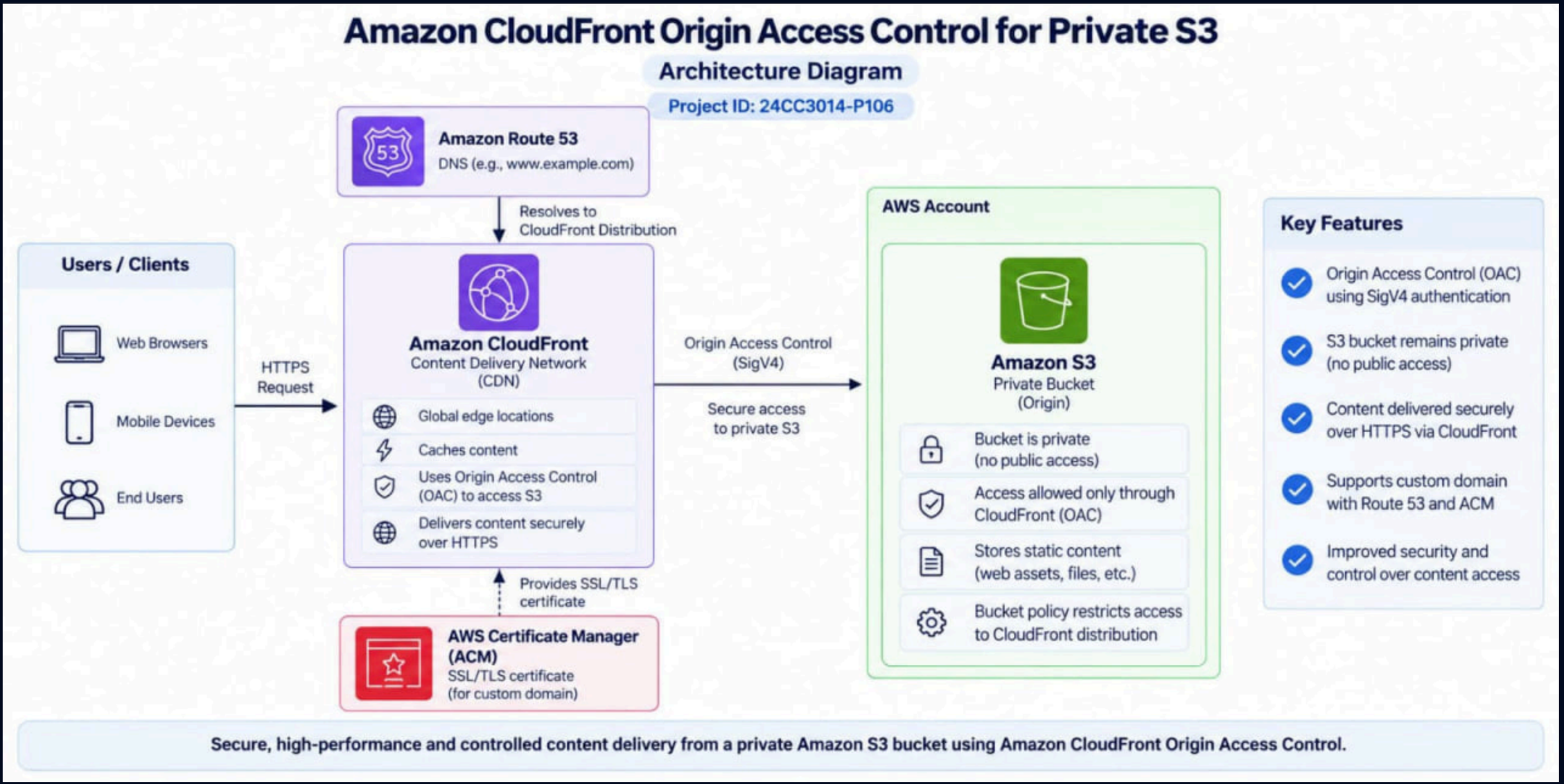
AWS SECURITY ARCHITECTURE

SOLUTION:

We propose a secure content delivery architecture using Amazon CloudFront and a private Amazon S3 bucket.

- Store website/static content in a private S3 bucket.
- Use Amazon CloudFront as the only public entry point.
- Configure Origin Access Control (OAC) to allow CloudFront to securely access S3.
- Use AWS Signature Version 4 (SigV4) to authenticate CloudFront → S3 requests.
- Use SSE-KMS to encrypt stored S3 objects and protect data at rest.
- Use HTTPS/ACM to secure data in transit.

ARCHTECTURE



Our Implementation

Roadmap to secure, private content delivery

PHASE 1
Secure private delivery

PHASE 2
Hardening and visibility



1. create S3 bucket
2. create CloudFront distribution with OAC
3. verify bucket policy
4. enforce HTTP to HTTPS
5. enable SSE-KMS encryption
6. monitor and CloudTrail auditing

Everything built and tested on a live AWS account

Phase 1

Building the Private Delivery Path

1



Created a private S3 bucket

(deepak-cloudfront-oac demo-2026, us-east-1)

with Block Public Access on

2



Created a CloudFront distribution and attached an Origin Access Control (OAC) to the S3 origin

3



Checked the generated S3 bucket policy: it allows only the CloudFront service principal for this distribution

RESULT | policy created successfully

The Test

DIRECT S3 URL

deepak-cloudfront-oac demo-2026.s3.us-east-1.amazonaux.com/index.html



BLOCKED

ACCESS DENIED

The bucket is private.
Direct access is not allowed.



CLOUDFRONT URL

d33lohyabeqhf4.cloudfront.net/index.html



SERVED

CONTENT DELIVERED

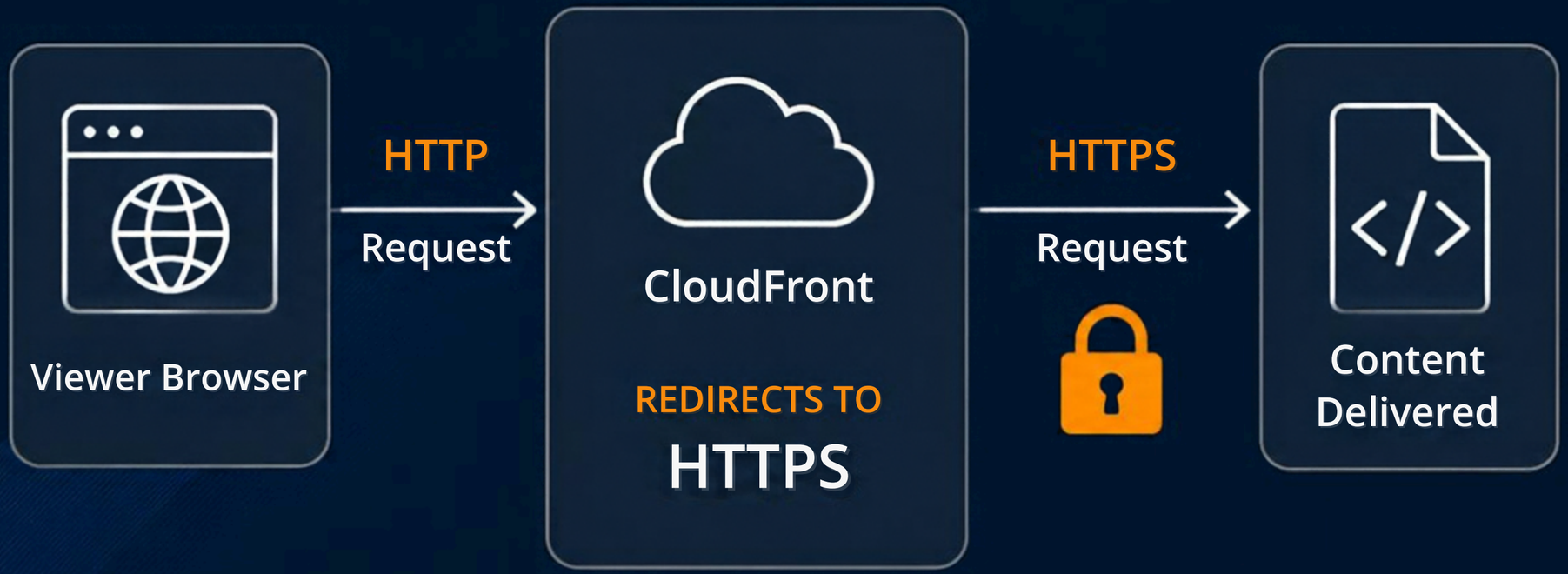
The only allowed path.
Content is delivered successfully.

SAME FILE, ONLY ONE ALLOWED PATH.

Proves OAC works: only CloudFront can read the bucket.

Phase 2, Step 4

Secure Delivery with HTTPS



Secure in Transit

Viewer traffic is encrypted with TLS.

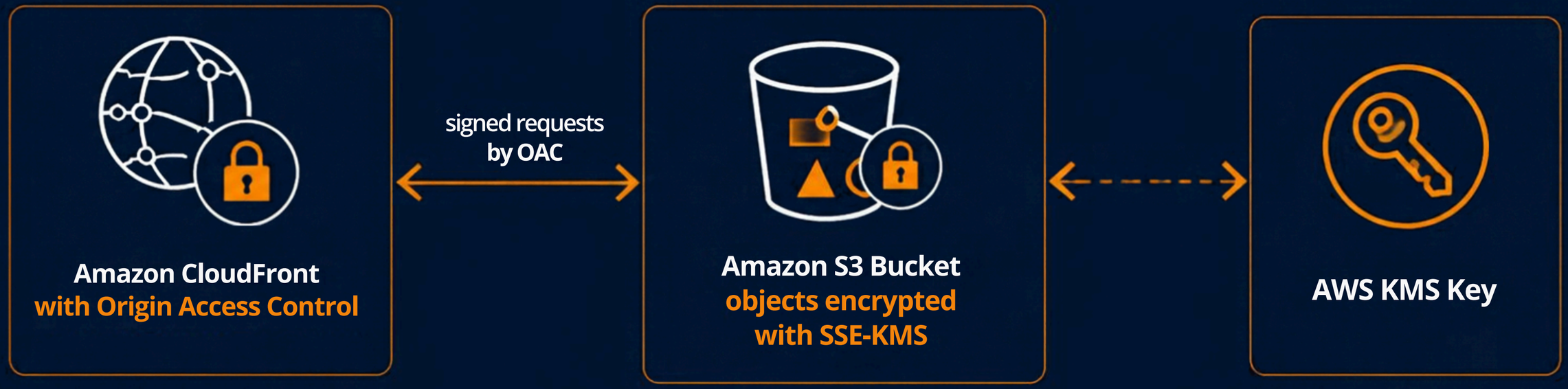
Trusted Delivery

HTTPS enforcement protects users and builds trust.

https://d33lohyabeqhf4.cloudfront.net/index.html

Phase 2, Step 5

SSE-KMS Encryption at Rest



LESSON LEARNED

The KMS key policy **must allow** the CloudFront service principal to decrypt, otherwise requests fail with 403

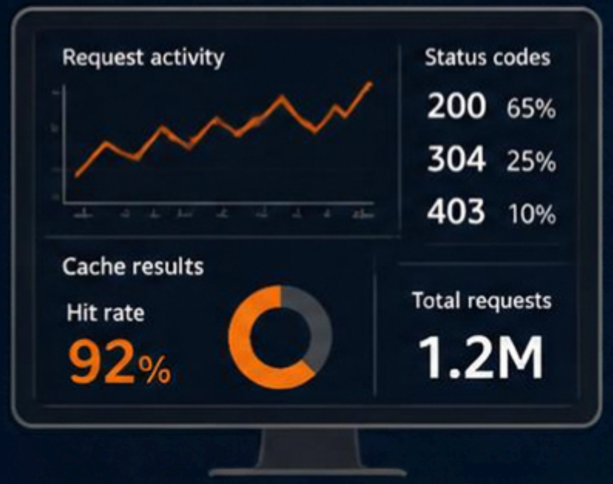
OAC supports SSE-KMS; the legacy OAI did not

Data is encrypted **at rest**, and traffic is encrypted **in transit**

Phase 2

Steps 5 and 6

5 Monitoring CloudFront requests



- Viewed request activity
- Checked status codes
- Verified cache results

6 AWS CloudTrail audit trail



- Created a trail that records account events
- Provides an audit history of who did what

1 Request
viewer request

→

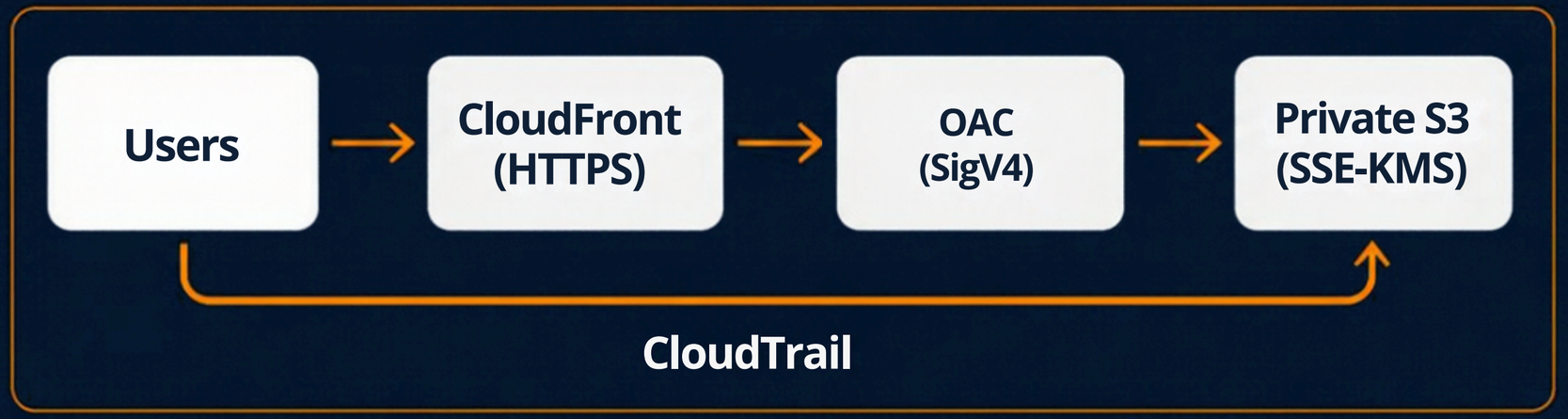
2 CloudFront logs
captures request activity

→

3 CloudTrail audit
records account events

What We Delivered

1	Private S3 bucket with no public access	✓
2	CloudFront distribution with OAC and SigV4	✓
3	Bucket policy scoped to the distribution	✓
4	HTTPS enforced for viewers	✓
5	SSE-KMS encryption at rest	✓
6	CloudFront monitoring and CloudTrail auditing	✓



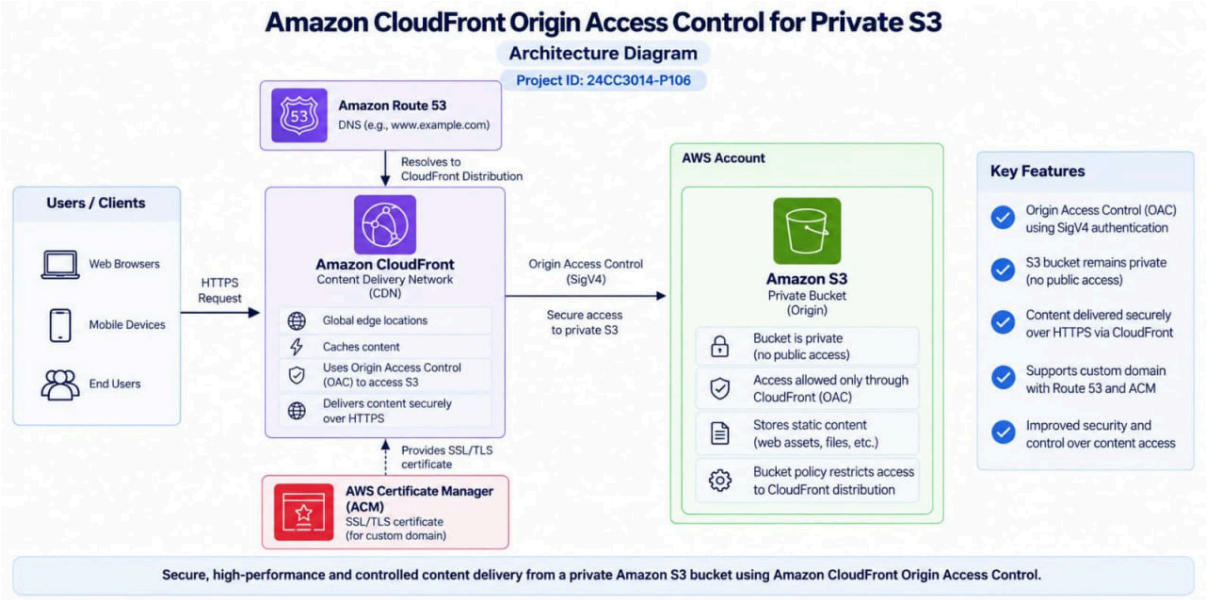
A private, encrypted, monitored content delivery pipeline built end to end.

DEMONSTRATION

AWS PROJECT

Problem statement:
Amazon CloudFront Origin Access Control for Private S3

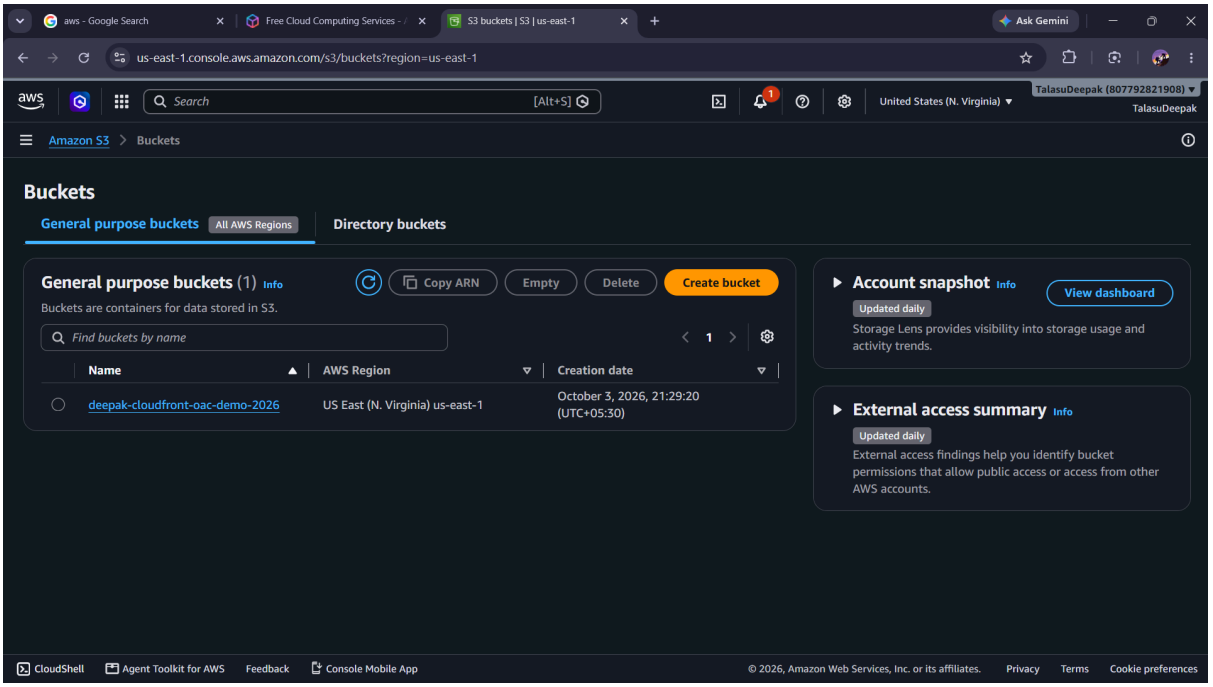
Architecture:



PHASE 1:

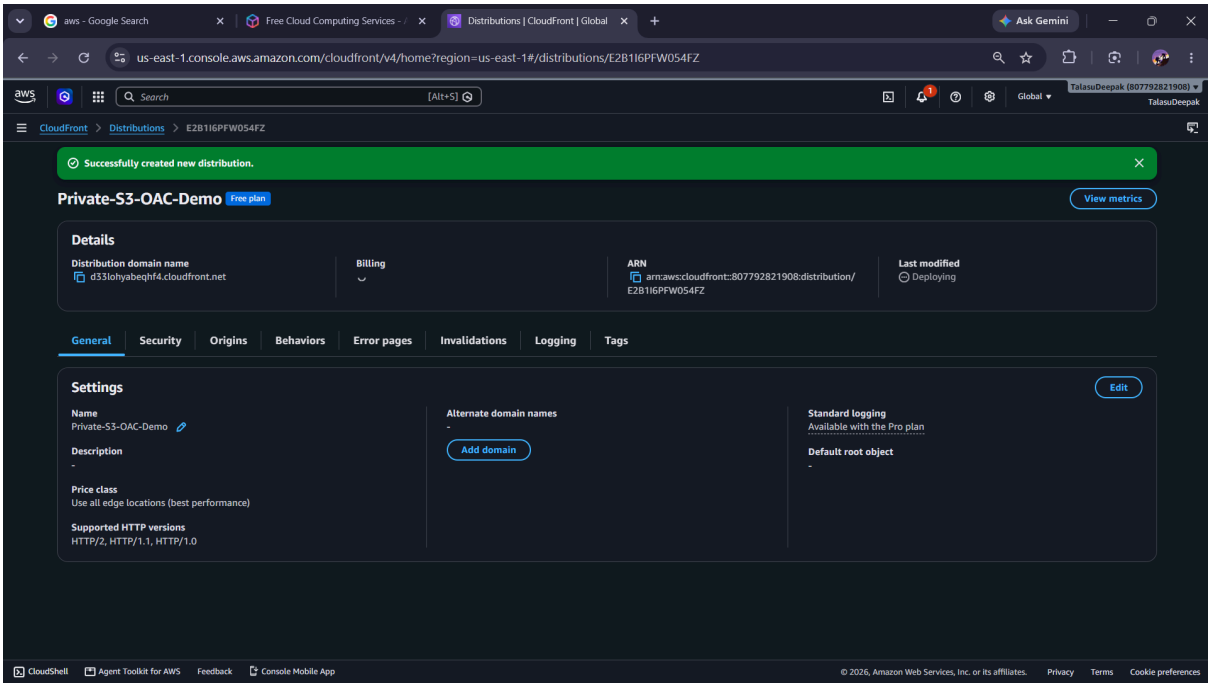
Step1:

Created an S3 bucket.

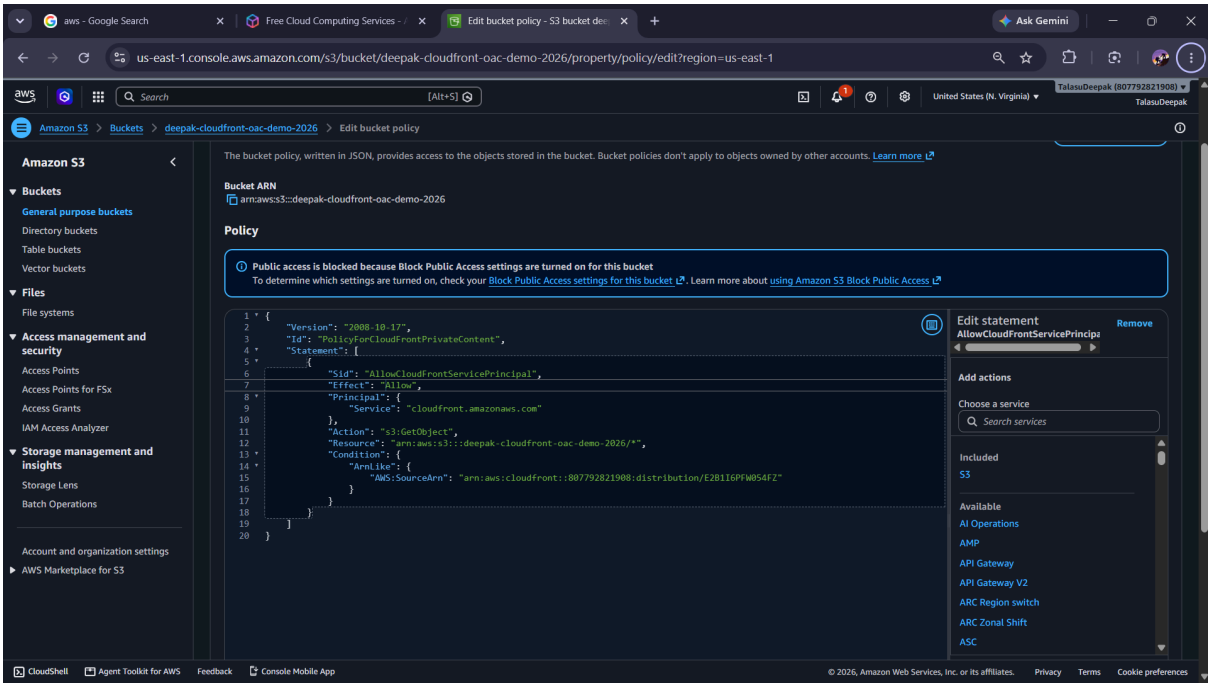


Step2:

Created a distribution i.e OAC in cloudfront:



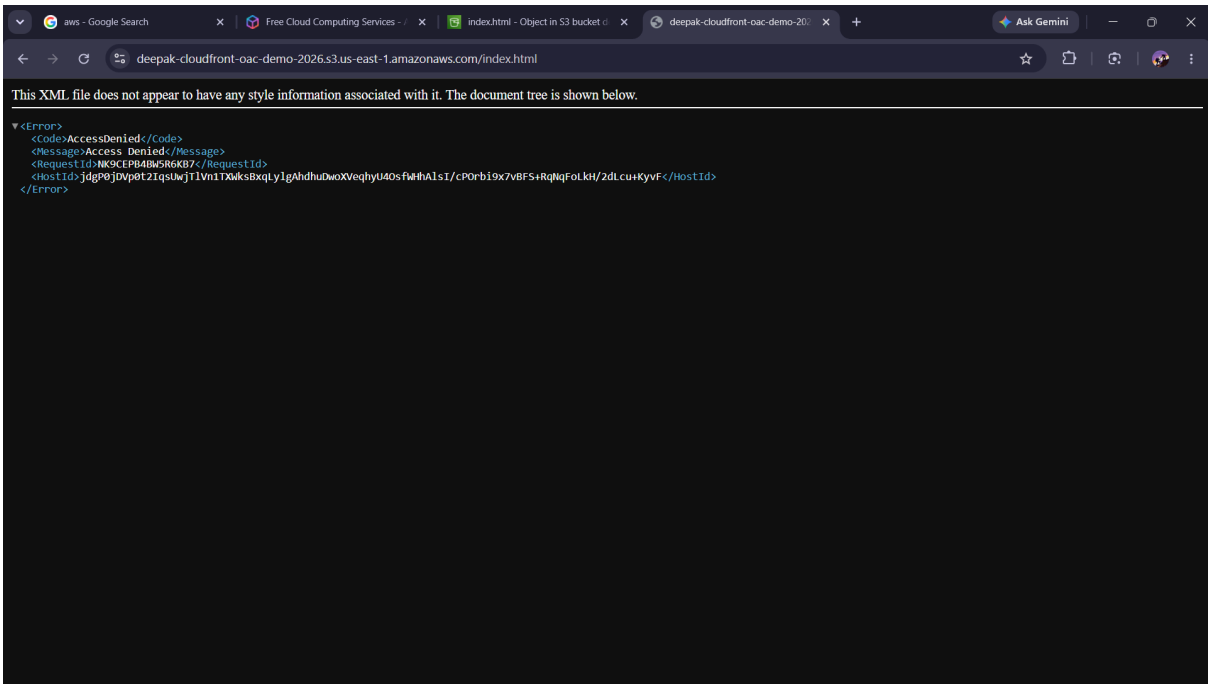
Step3:
Checked AWS bucket policy:



Created successfully

Now testing time:

Direct s3 object url:
http://deepak-cloudfront-oac-demo-2026.s3.us-east-1.amazonaws.com/index.html



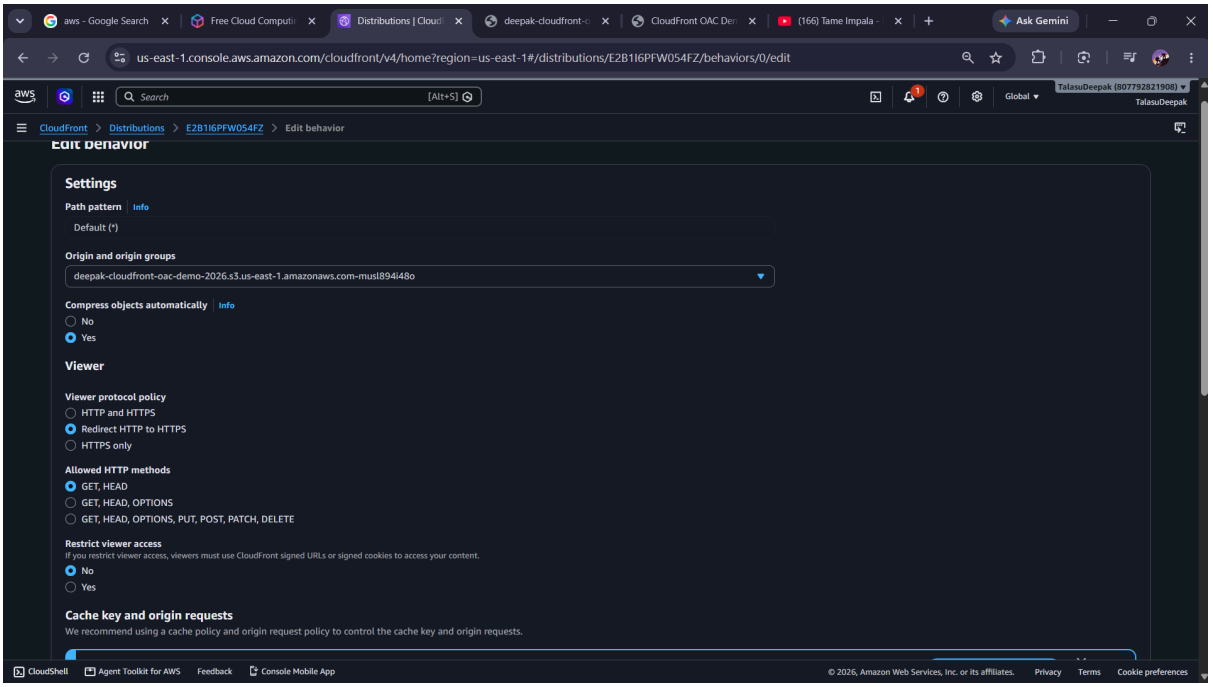
With cloudfront url:
http://d33lohyabeqhf4.cloudfront.net/index.html

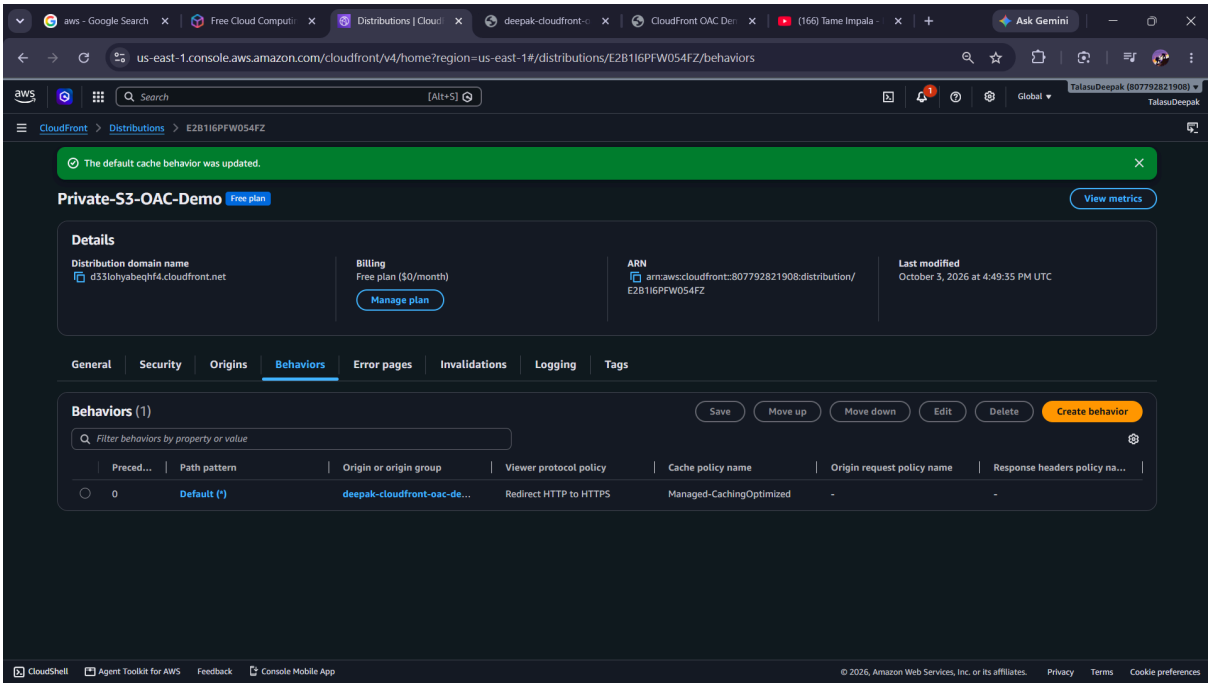
PHASE 1 completed…

PHASE 2:

Step4:

Configuring from http to https for secure requests…

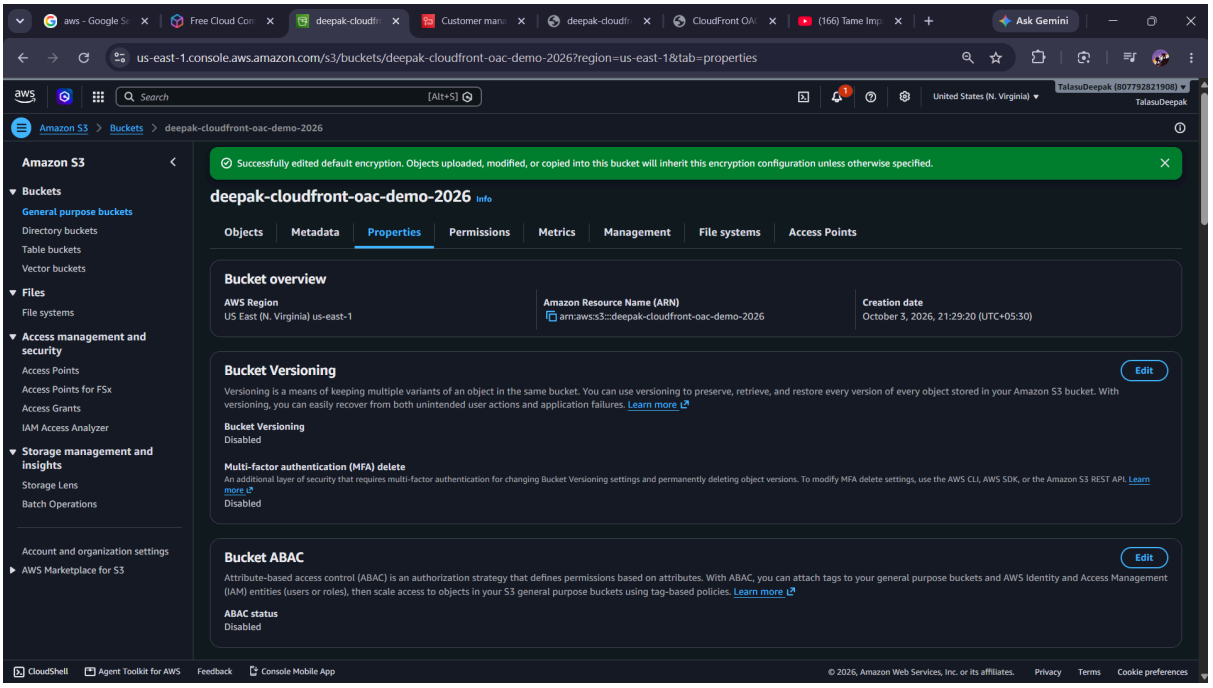


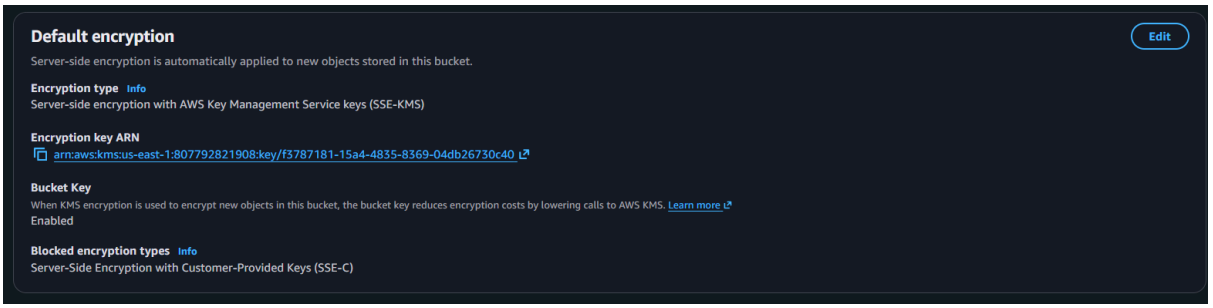


Link updated:
https://d33lohyabeqhf4.cloudfront.net/index.html

Step4:

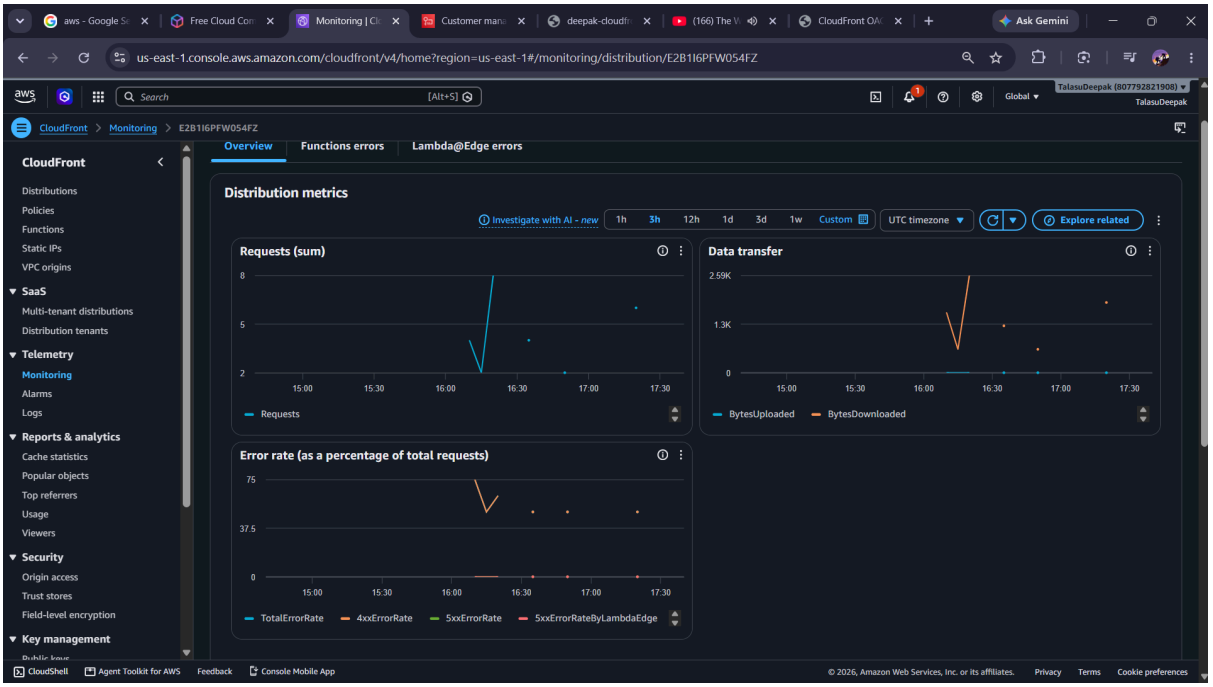
Implemented SSE - KMS ENCRYPTION…





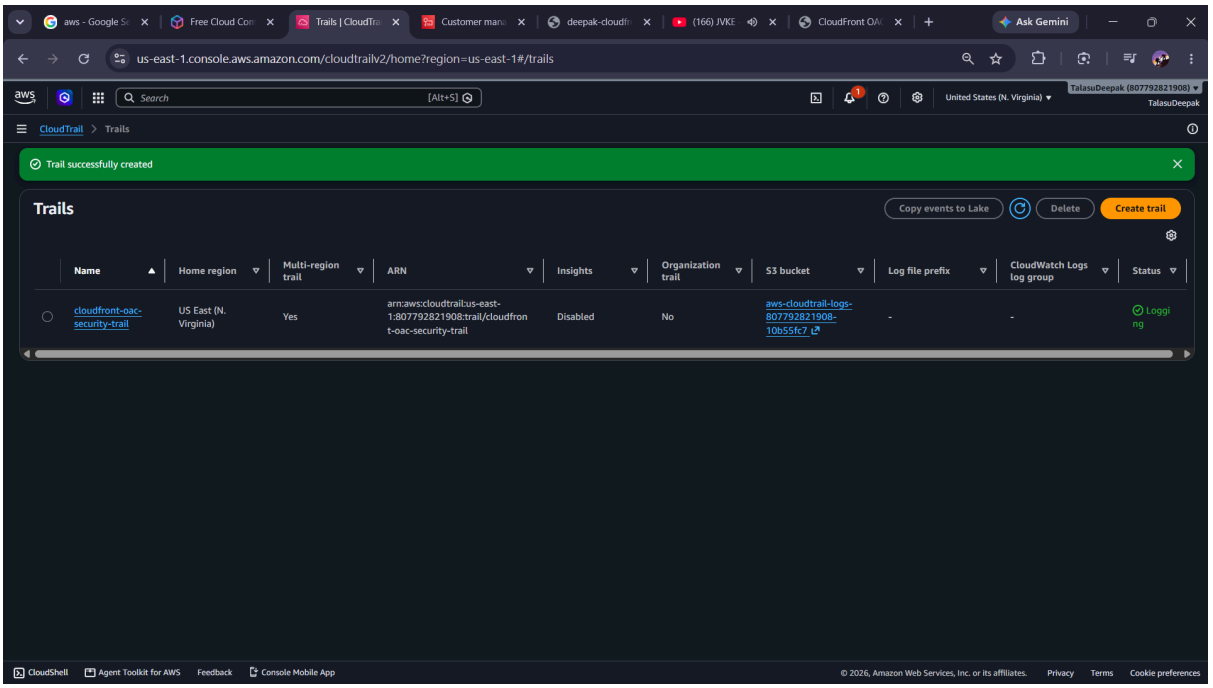
Step5:

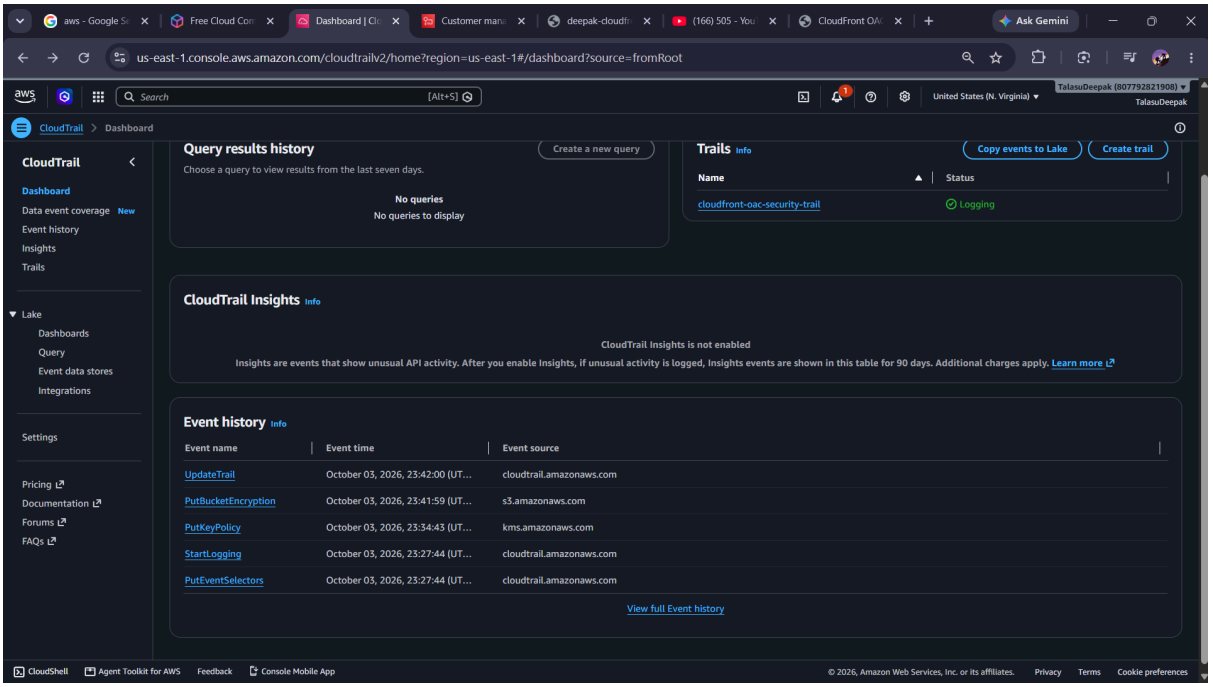
Monitoring cloud front requests:



Step6:

Created a cloudtrail:





Showing what events have been occurred

